Sukhrob Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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Research Interests

Vision models that clinicians and scientists can trust enough to deploy. Three questions run through my work: can a clinician check what a model attended to, instead of taking a score on faith; how much of a black-box generative model can be controlled from outside, without retraining; and do learned representations hold up on scientific imaging, where data looks nothing like ImageNet and labels are noisy. I also build ML for a HIPAA-regulated clinical platform, where the same questions have patients behind them.

Education

Northeastern University, Boston, MA *January 2024 – May 2026*

- Master of Science in Computer Science; GPA 4.0/4.0
- Coursework: Advanced Perception, Computer Vision, Machine Learning

Suffolk University, Boston, MA *September 2021 – May 2023*

- Bachelor of Science in Computer Science; Cum Laude; GPA 3.5/4.0
- Transferred from INTI International College, Malaysia (2019 – 2021) and Tashkent University of Information Technologies, Uzbekistan (2017 – 2019)

Publications

[1] **S. Ilyosbekov**, S. Gajjar, and R. Jin, "MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images," *arXiv preprint*, arXiv:2608.22690, 2026. [Paper]

[2] **S. Ilyosbekov**, "Localize, Don't Beautify: Client-Side Control of Image-Editing APIs for Cosmetic Surgery Previews," *arXiv preprint*, arXiv:2608.02841, 2026. [Paper]

[3] **S. Ilyosbekov**, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2025. [Paper]

Awards & Honors

- **Associate Member,** Sigma Xi, The Scientific Research Honor Society, 2026
- **Outstanding Project Award,** Bookshelf Scanner, Computer Vision course, Northeastern University
- **National Physics Olympiad, 1st Place,** Uzbekistan; first student from my high school to win nationally
- **University Coding Competition Winner,** Tashkent University of Information Technologies

Research

MorphoCLIP: Text-Supervised Cell Painting Retrieval GitHub | Paper

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Contrastive model matching Cell Painting images of drug- and gene-perturbed cells to plain-language treatment descriptions.
- Frozen DINOv3 and BioClinical ModernBERT backbones with trained projection heads; trains on one consumer GPU.
- Bidirectional retrieval on CPJUMP1: 51 plates, 3M+ images, 303 compounds, 160 genes. Embedding-space batch correction against plate effects.
- First author with S. Gajjar and R. Jin; led model design, training, and evaluation.

Localize, Don't Beautify: Client-Side Control of Image-Editing APIs Paper

Image-editing APIs, inpainting models, ArcFace, prompt engineering

- How far a black-box image-editing API can be controlled client-side, with cosmetic-surgery previews as the test case.

- Prompt steering, masked compositing, and model-based inpainting compared across six commercial editors on 196 facelift and rhinoplasty edits.
- Scored on identity preservation (ArcFace) and edit localization; masked compositing localized best. Sole author.

MelanomaNet: Explainable Deep Learning for Skin Lesion Classification GitHub | Paper

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- EfficientNet V2 at 384×384 with focal loss over all nine ISIC 2019 categories; 85.6% accuracy and 0.856 weighted F1 on 25,331 dermoscopic images.
- GradCAM++ attention decomposed along the ABCDE criteria, with asymmetry, border irregularity, color variation (K-means), and diameter measured from the lesion mask.
- Agreement between clinical features and model attention is scored numerically, not by example heatmaps. Sole author.

Selected Course Projects

LightDepth: Lightweight Monocular Depth Estimation GitHub

PyTorch, ResNet18, U-Net, NYU Depth V2

- ResNet18 encoder with a U-Net decoder; 14.3M parameters against 24.8M for Depth Anything V2, 42% fewer, 72% faster inference.
- On NYU Depth V2, AbsRel 5.3% lower and SqRel 31.1% lower relative to that baseline.

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network GitHub

PyTorch, mixed-precision training, T91, Set5, Set14, DIV2K

- Reproduced FSRCNN (Dong et al., ECCV 2016) at 2×, 3×, and 4× with end-to-end learned upsampling.
- Matched the paper's gains over SRCNN: +1.78 dB PSNR on Set5, +1.26 dB on Set14. Added ablations on the shrinking and mapping layers.

Bookshelf Scanner: Multi-Modal Book Detection and Recognition GitHub

YOLO, Moondream2 VLM, llama.cpp, PyTorch, FastAPI

- YOLO instance segmentation isolates each book spine; Moondream2 reads the title and author off it. Course Outstanding Project Award.

Work Experience

Senior Software Engineer | EmTech Care Labs | Portland, ME *Jan 2025 – Present*

- Decline-risk model over daily activity and care-log data for Alzheimer's residents; 0.81 AUC on held-out data.
- NLP pipeline extracting symptoms, medications, and care events from clinical notes; 0.90 F1. Serving on de-identified data under HIPAA.

Senior Software Engineer | AllFactors | Santa Clara, CA *Oct 2021 – Dec 2024*

- Property-valuation and market-forecast models over 2M listings in 40+ metros; median error 9% to 5% against the rules-based baseline.
- Real-time feature pipeline on SignalR and SQL Server change tracking; predictions served in Blazor dashboards.

Earlier: Software Engineer, Smart Meal Service, Russia (2020 – 2021). Game Developer, Pentalight Technology, Malaysia (2020 – 2021). Freelance Developer (2019 – 2020). Game Developer, EC Dev Team, Uzbekistan (2016 – 2019).

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, CNNs, Vision Transformers, DINOv3, U-Net, EfficientNet, YOLO, contrastive and self-supervised learning, GradCAM++, detection and segmentation, depth estimation

Mathematics: Linear algebra, probability and statistics, convex optimization, numerical methods, information theory, hypothesis testing and experimental design

Machine Learning: Transfer learning, loss design, regularization, hyperparameter search, cross-validation and error analysis, XGBoost, K-means, NLP

Research Computing: Python, NumPy/SciPy, pandas, scikit-learn, Hugging Face, CUDA, mixed-precision and multi-GPU training, Weights & Biases, Docker, LaTeX

Software Engineering: C#, C++, TypeScript; .NET, FastAPI, PostgreSQL, Kubernetes, AWS, Azure